

Optimization-Based Clinical Decision Support for Curable Hypertension

SVMs: Hard-Margin Feasibility, Soft Margins, and Kernel Validation

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The story in one line

strict $\rightarrow$ prove impossible $\rightarrow$ relax $\rightarrow$ compare $\rightarrow$ conclude

- Try the **hard-margin** SVM: a straight line with zero mistakes.
- **Prove** no such line exists — three independent ways.
- **Relax** to a soft margin that charges for mistakes.
- **Compare** against a curved (RBF kernel) boundary.
- **Conclude**: the simple linear model is the right choice.

The five steps — what each one asks and answers

- 1 Hard margin — can a perfect line exist?** Demand a straight boundary that classifies every patient correctly with a margin. This is a convex feasibility problem: either such a line exists, or it provably does not.
- 2 Prove it is impossible.** Three independent checks agree the classes are not linearly separable: (2a) the LP solver returns “infeasible”; (2b) a 12-patient Farkas certificate turns that into a proof ($0 \geq 1$); (2c) the two convex hulls overlap.
- 3 Relax to the soft margin.** Allow mistakes but charge for them through slack variables and a penalty C . Always solvable; (3a) tune C by grid search, (3b) read it as a Pareto trade-off, (3c) see the fitted line, (3d) solve the primal directly by SGD.
- 4 Compare against a curved boundary.** Swap the inner product for an RBF kernel so the boundary can bend, then check whether the extra flexibility actually helps or merely overfits.

Takeaway: Steps 1–2 establish *what is impossible*; steps 3–4 find *the best thing that is possible*.

The clinical problem

Primary aldosteronism (PA) is a common, *curable* cause of high blood pressure.

- **Unilateral (UPA)**: one gland — often *cured by surgery*.
- **Bilateral (BPA)**: both glands — managed with *medication*.

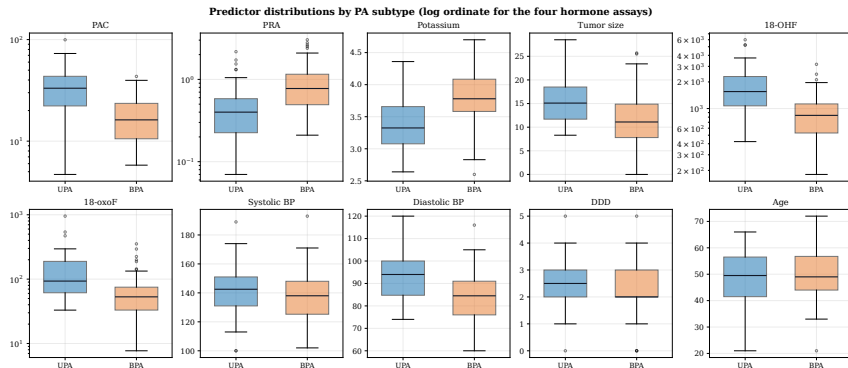
Getting the subtype right changes the treatment.
Can an optimization model separate the two from routine measurements?

Takeaway: Frame subtype classification as a maximum-margin optimization problem.

The dataset

- 114 synthetic patients
- 56 UPA / 58 BPA
- 10 continuous predictors (PAC, PRA, potassium, tumor size, steroids, BP, ...)

What the data look like before any model



114 patients, no missing values. **UPA** versus **BPA**; hormone assays on a log ordinate.

Takeaway: Every panel overlaps — no single measurement splits the two subtypes.

Descriptive statistics — the subtype contrast

Feature	UPA	BPA	Cohen's d	Welch t	p
PAC (ng/dL)	34.9	18.4	1.17	6.20	<0.001
18-OHF (ng/dL)	1883	942	0.99	5.25	<0.001
Potassium (mmol/L)	3.38	3.79	-0.95	-5.09	<0.001
PRA (ng/mL/h)	0.51	0.96	-0.80	-4.32	<0.001
Diastolic BP (mmHg)	92.3	84.1	0.75	4.03	<0.001
Tumor size (mm)	15.4	11.4	0.74	3.98	<0.001
18-oxoF (ng/dL)	148.1	71.8	0.66	3.46	<0.001
DDD (drugs)	2.48	2.14	0.36	1.94	0.055
Systolic BP (mmHg)	141.4	137.5	0.23	1.22	0.224
Age (years)	48.6	50.1	-0.15	-0.80	0.425

Bold: $p < 0.05$ (Welch t -test), i.e. separates the subtypes (7 of 10). Sign: + means higher in UPA.

Cohen's d — how big

Gap between *class means*, in pooled SDs: 0.2 small, 0.5 medium, 0.8+ large.

Welch t — how certain

Same gap $\div$ standard error; grows with $\sqrt{n}$. Evidence, not size.

Takeaway: Strong univariate signal, but overlapping distributions — a perfect separator is in doubt.

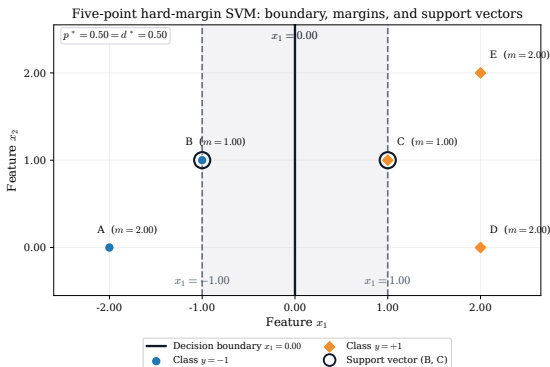
Step 1 — the strictest model: hard-margin SVM

Find the **widest empty corridor** between the two classes:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1.$$

- Score $s = w^\top x + b$; sign is the prediction.
- Margin width $= 2/\|w\|$.
- **Zero mistakes allowed.**

A convex quadratic program — *if* a feasible line exists.



Five-point teaching example: boundary, margins, support vectors.

Step 2 — prove no such line exists

We attack the question **three independent ways**. If any one succeeds, the hard-margin model is impossible — and all three agree.

2a. Ask the solver

Run the feasibility LP on all data and every fold.

2b. Farkas certificate

Get a signed receipt — 12 patients that *prove* $0 \geq 1$.

2c. Convex hulls

Show the two classes literally share a point.

Each is stronger than the last: a solver *says* no; Farkas *proves* no; the hulls *show* why.

Step 2a — ask the solver

Feasibility of the hard-margin constraints is itself a **linear program**: does any (w, b) satisfy

$$t_i(w^\top x_i + b) \geq 1 \quad \text{for all } i?$$

- HiGHS returns **status: infeasible** on the full 114-patient dataset.
- Repeated inside every one of the 5 cross-validation folds — **all infeasible**.
- So the result is not an artifact of one particular split.

Takeaway: A “no” from the solver is a starting point — but why should we trust it? Steps 2b–2c give proof.

Solver verdict

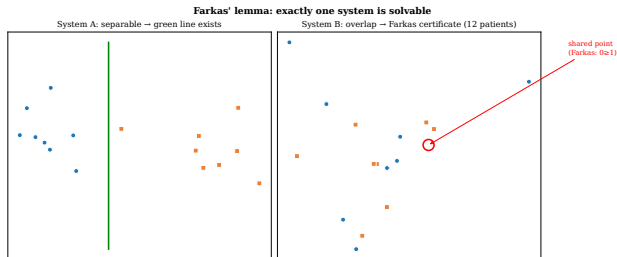
Subset	Feasible?
Full data (114)	No
Fold 1	No
Fold 2	No
Fold 3	No
Fold 4	No
Fold 5	No

Step 2b — the Farkas certificate

Farkas' lemma: exactly *one* of two systems can hold — either a separating w exists, **or** a set of nonnegative weights λ gives a contradiction.

- The solver hands back **12 patients** with weights $\lambda_i \geq 0$.
- They combine to prove $0 \geq 1$ — impossible.
- This is a *checkable receipt*: no need to trust the solver's search.

Takeaway: The certificate turns “the solver said no” into a proof anyone can verify by hand.



Only one column can be true. On this data it is the right one.

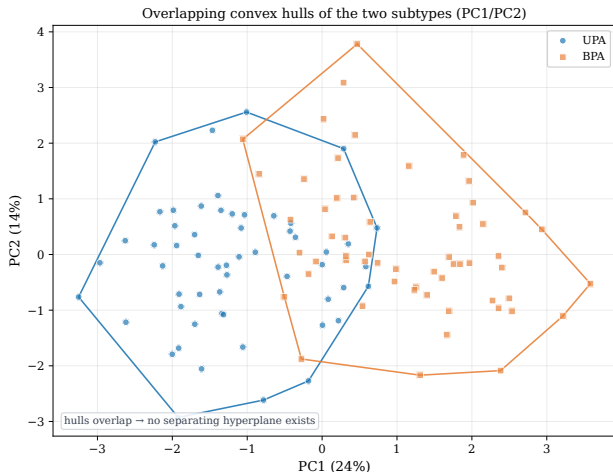
Step 2c — what the certificate means: overlapping hulls

The same λ weights have a picture.
Rescaled, they build **one point inside both classes' convex hulls**:

$$\sum_i \beta_i^+ x_i = \sum_i \beta_i^- x_i,$$

matching to 6.66×10^{-16} (machine zero).

- A shared point $\Rightarrow$ no hyperplane separates the classes.
- Geometry, solver, and algebra tell one story.



The real cohort in a 2-D projection: the hulls clearly overlap. The full-dimensional proof is exact.

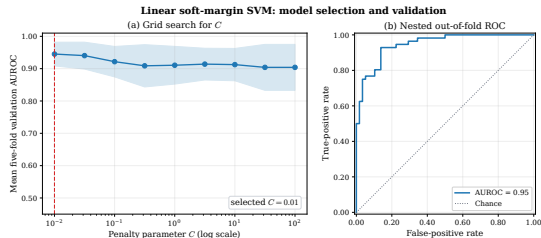
Step 3 — relax: the soft-margin SVM

Allow mistakes, but **charge** for them with slack ξ_i :

$$\min_{w, b, \xi} \quad \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$$\text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

- C = strictness knob (chosen by nested CV).
- Selected $C = 0.01$ (leakage-safe grid search).
- Always solvable — overlap is modeled, not assumed away.



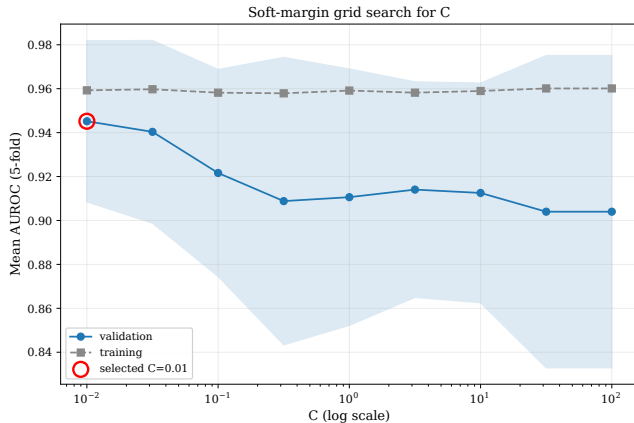
(a) validation AUROC vs C ; (b) nested out-of-fold ROC, **AUROC 0.95**.

Step 3a — choosing C by grid search

Sweep C over $10^{-2} \dots 10^2$ and score each by **inner-fold validation** AUROC (never the test fold — leakage-safe).

- Training AUROC stays flat near 0.96 at every C .
- Validation AUROC *peaks at the smallest* $C = 0.01$.
- Larger C (harder penalty) $\Rightarrow$ worse generalization.

The gap between the two curves *is* the overfitting.



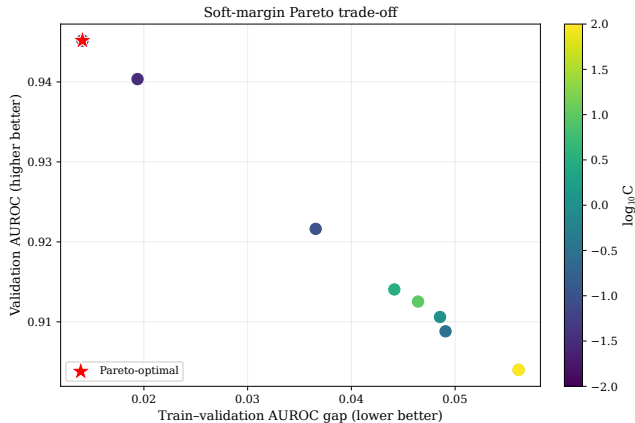
Mean validation vs training AUROC across the C grid; band is ± 1 SD over folds. The selected $C = 0.01$ is circled.

Takeaway: Stronger regularization (small C) generalizes best on this overlapping data.

Step 3b — the same grid as a Pareto trade-off

Plot each C as (train-validation gap, validation AUROC). The best model is **top-left**: high AUROC *and* little overfitting.

- $C = 0.01$ is the **single Pareto-optimal** point — it wins on *both* axes at once.
- Every larger C gives away AUROC *and* widens the gap.
- No trade-off to negotiate: the simplest penalty is strictly best here.



Color encodes $\log_{10} C$; the star is the selected, Pareto-optimal model.

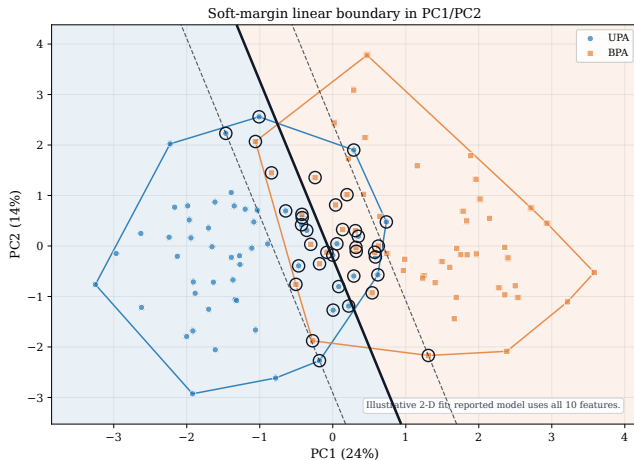
Takeaway: $C = 0.01$ dominates the grid on both axes at once.

Step 3c — what the soft-margin line looks like

Projecting the cohort onto its first two principal components makes the fitted **soft-margin boundary** visible.

- The line runs *straight through the overlap* where the two hulls intersect.
- Circled points are **support vectors** — almost all sit in the contested band.
- Points inside the margin or on the wrong side are exactly what the slack ξ_i pays for.

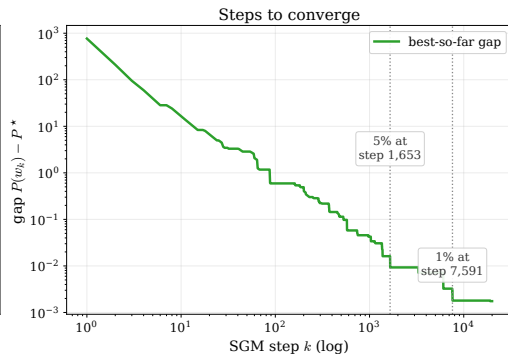
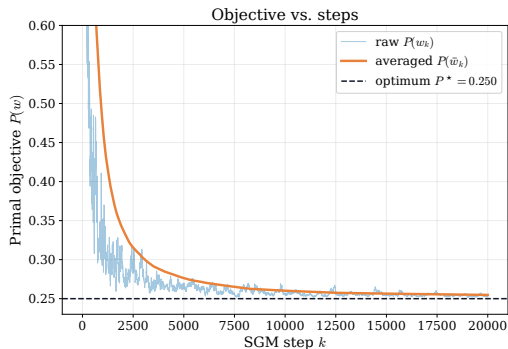
This 2-D fit is illustrative; the reported model uses all 10 features.



Solid: decision boundary; dashed: the ± 1 margins. Hulls overlap, so no line separates them cleanly.

Step 3d — solving the primal directly with SGD

The soft-margin primal is a **finite sum** over the m patients, so it can be solved one random point at a time (no dual, no QP): $w_{k+1} = w_k - \eta_k g_k$ with $g_k = \lambda w_k$ if $t_k w_k^\top x_k \geq 1$, else $g_k = \lambda w_k - t_k x_k$ (step $\eta_k = 1/(\lambda k)$; the hinge kink makes g_k a *subgradient*, as in the lecture, plus the regularizer term). Reaches within 5% of the optimum in ≈ 14 epochs, 1% in ≈ 67 , and matches the dual solver (cosine 0.99, AUROC 0.958 vs 0.955).



Takeaway: SGD reaches the same hyperplane as the dual — but the noisy floor is a tell: the data are not separable ($B > 0$).

Step 4 — from soft margin to a kernel

1. **The soft-margin primal** (what we solved in Step 3):

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad t_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

2. **Its Lagrangian dual** — the data enters **only through inner products** $x_i^\top x_j$:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j t_i t_j \underbrace{x_i^\top x_j}_{\text{inner product}} \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i t_i = 0.$$

3. **The kernel trick.** Map $x \mapsto \varphi(x)$ into a higher space and replace every inner product by a **kernel**:

$$x_i^\top x_j \longrightarrow K(x_i, x_j) = \varphi(x_i)^\top \varphi(x_j).$$

4. **Choose the RBF kernel:**

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2).$$

A curved boundary in the original space, still a convex QP in α .

$$\text{Decision score becomes } f(x) = \sum_i \alpha_i t_i K(x_i, x) + b.$$

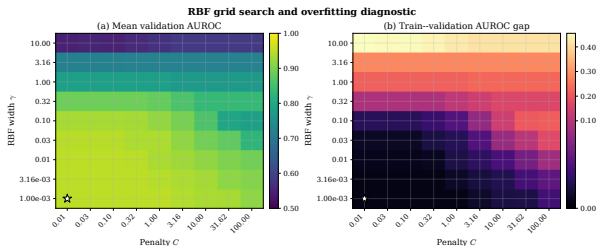
Step 4 — compare against a curved boundary (RBF kernel)

The RBF kernel

$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$ can bend the boundary.

- Joint search over 81 (C, γ) pairs, nested CV.
- Best: $C = 0.01$, $\gamma = 0.001$ — the *smoothest* setting.
- Large γ : perfect training, near-chance validation (overfit).

The data wanted a nearly-straight boundary.

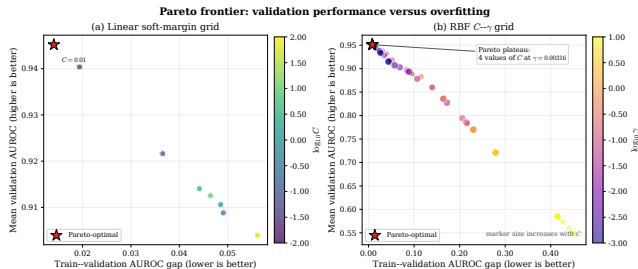


Validation AUROC (left) vs train-validation gap (right).
The star marks the selected smooth model; the top rows overfit.

Step 4 — performance versus overfitting: the Pareto view

Plot every candidate as (train-validation gap, validation AUROC). The **best models sit in the top-left**: high AUROC, small gap.

- **Linear**: the whole frontier is tightly clustered top-left — little room to overfit.
- **RBF**: a long tail trades generalization for training fit as γ grows.
- The Pareto-optimal RBF point coincides with the *smoothest* setting — essentially the linear model.



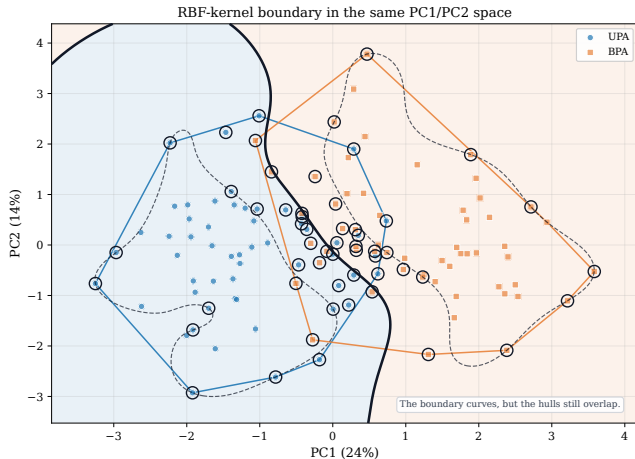
The star is the Pareto-optimal model in each grid. RBF's extra flexibility only buys a longer overfitting tail, not a better frontier.

Step 4 — the kernel boundary redrawn in PC1/PC2

The same projection as Step 3c, now with the **RBF-kernel** decision surface.

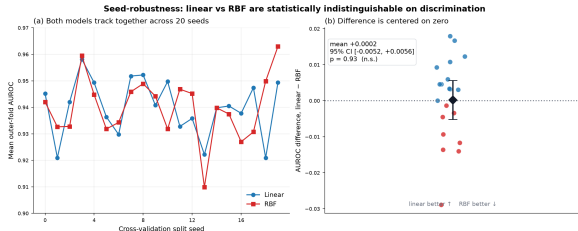
- The boundary is **curved**, not straight — the kernel's extra flexibility at work.
- It bends *around* individual points, yet the two hulls still overlap.
- Curvature relabels a few borderline points but does not carve out a clean separation.

Illustrative 2-D fit (moderate γ); the tuned model chose a much smoother γ .



Compare with the straight line of Step 3c: more shape, same overlap.

Are linear and RBF actually different?



Fair metric (mean outer-fold AUROC), 20 fold seeds:

- Difference = +0.0002
- 95% CI [-0.005, +0.006]
- paired t : $p = 0.93$ (n.s.)
- Linear $\geq$ RBF in 12/20 seeds.

Statistically indistinguishable on discrimination.

Takeaway: The kernel never improves on the line; the tiny gap is just the random split.

Step 5 — conclude

Verdict

- Hard-margin → **impossible** (proven three ways).
- Linear soft-margin → **AUROC 0.95** ✓
- RBF kernel → AUROC 0.94 (no gain).

The simpler model wins.

Fewer tuning knobs, an interpretable score, equal discrimination — parsimony decides.

Why it matters

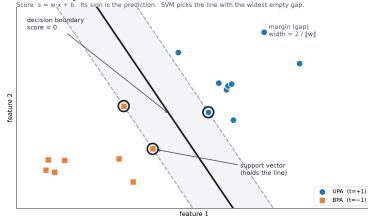
- The negative feasibility result is *informative*, not a bug: clinical measurements genuinely overlap.
- Complexity must be justified by *validation*, never by training fit.
- Synthetic data $\Rightarrow$ methodological, not clinical, claim.

Support Vector Machines — One-Page Cheat-Sheet

strict $\rightarrow$ prove impossible $\rightarrow$ relax $\rightarrow$ compare $\rightarrow$ conclude

A. Anatomy: what an SVM draws

Score $s = w \cdot x + b$. Its sign is the prediction. SVM picks the line with the widest empty gap.



B. The report's logic, step by step

1. Try the STRICTEST model

Hard-margin: a straight line with ZERO mistakes allowed.

2. PROVE it's impossible

3 ways agree: solver says 'infeasible', a 12-patient Farfars receipt gives $0 \geq 1$, and the two groups overlap exactly.

3. RELAX just enough

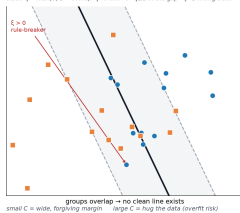
Soft-margin: allow mistakes, but charge for them (knob C). Always solvable. $\rightarrow$ held-out AUROC = 0.95 ✓ winner

4. COMPARE vs. curved

RBF kernel (knob γ). Tuning chose the smoothest γ ; AUROC 0.94 — no better. High γ overfits badly: $\rightarrow$ keep it simple.

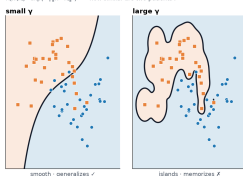
C. Soft-margin: mistakes cost money

slack $\xi = \max(0, 1 - \text{score})$. $\xi=0$ fine. $0 < \xi \leq 1$ in the gap. $\xi > 1$ wrong side.



D. The γ knob (RBF): smooth vs. memorize

$K(x, x) = \exp(-\gamma \|x - x_j\|^2)$ — 'how similar are two patients'.



Here tuning chose the smoothest γ — the data wanted a nearly-straight boundary.

E. Symbol key & verdict

x	the 10 measurements for a patient
$t = \pm 1$	the answer: +1 UPA, -1 BPA
w	direction of the dividing line
b	offset — shifts the line
$w \cdot x + b$	the score (sign = prediction)
$\ w\ $	length of w : shorter = wider gap
a	how hard a patient pulls the line
ξ	a patient's rule-breaking amount
C	strictness knob for mistakes
γ	curviness knob (RBF only)

VERDICT

Hard-margin $\rightarrow$ impossible (proven)

Linear soft-margin $\rightarrow$ AUROC 0.95 ✓

RBF kernel $\rightarrow$ AUROC 0.94 (no gain)

Simpler model wins.

Thank you

Questions?

Reproduced end-to-end: 30/30 result files match; independent nested-CV and statistical tests confirm the verdict.